Octo · arXiv 2405.12213 · RSS 2024

Scientific Reviewer

8+3 · seat 1 = Kanta · HIGH

12 critiques + reporting checklist
Open FT ancestor — not BEHAVIOR drop-in

Oct 14 paper club · VLA Architectures · Oct 16 BEHAVIOR nearby

Spine: open OXE GRP + diffusion chunks · underspecified horizons · NOT 2026 winner weights

Octo Model Team · Berkeley / Stanford / CMU / DeepMind · HF rail-berkeley/octo-*

Reviewer talk arc · 8 minutes

1	2	3	4	5
Steelman	**12 critiques**	**Checklist**	**Cite / verdict**	**Q&A**
60s	3.5 min	90s	90s	3 min
Open + adapters + chunks + FT	THE figure Say aloud	72% / 29% 83% / 5%	Ancestor ≠ BEHAVIOR drop-in	Horizons RT-2-X / TE

GitHub title: [ECE 605] - Octo Making Scientific Reviewer

Spine: open OXE GRP + diffusion chunks · underspecified horizons · NOT 2026 winner / NOT BEHAVIOR drop-in

Steelman · what the paper actually delivers

Open GRP stack

Weights + train + loaders
Small/Base ckpts · HF
RT-2-X / RoboCat did not match

Compositional I/O

Block-wise attention
Mask missing cams/lang
Add lightweight adapters at FT

Diffusion chunks

ViT-first + DDPM head
Imports ACT/DP control culture
Table II: diffusion ≫ MSE/discrete

Operational FT recipe

~100 demos · 50k steps
~5 h on A5000 24 GB
Shared HPs across 6 domains

Integrative + open systems novelty — not a single new module

Cite openness + finetune API as the primary artifact,
not “beats RT-2-X” alone. Component pieces (transformer, OXE, diffusion, chunks) each existed.

Octo stack · ViT + frozen T5 + DDPM chunk head

Single backbone forward · light head denoises · receding-horizon execute (TE not helpful)

Task tokens

Language → frozen T5-base (111M) → 16 emb tokens
—or— Goal image CNN patches (hindsight relabel)

Obs tokens

Primary 256^2 → 256 patches
Wrist 128^2 → 64 patches
History H=2 · block mask
Missing modalities masked

Octo Transformer (ViT)

Small 27M · D=384 · 12L · 6H

Base 93M · D=768 · 12L · 12H

Block-wise causal attention

Readout tokens (CLS-like) attend → not attended into → passive emb for action head

FT: add encoders / heads w/o re-init backbone

DDPM action head

3-layer MLP · hid 256
~20 diffusion steps · cosine
Predict action chunk $\hat{a}_{t:t+H}$

Pretrain: ΔEE + abs gripper

Deploy control

Execute H_{exec} prefix → replan
TE: no extra benefit (App. E)
Bimanual ex: pred 64 / exec 12
Log commit_seconds = H_{exec}/Hz

Club control one-liner

Cross-embodiment receding-horizon diffusion-chunk controller — not ACT TE by default · not a 2026 VLM winner

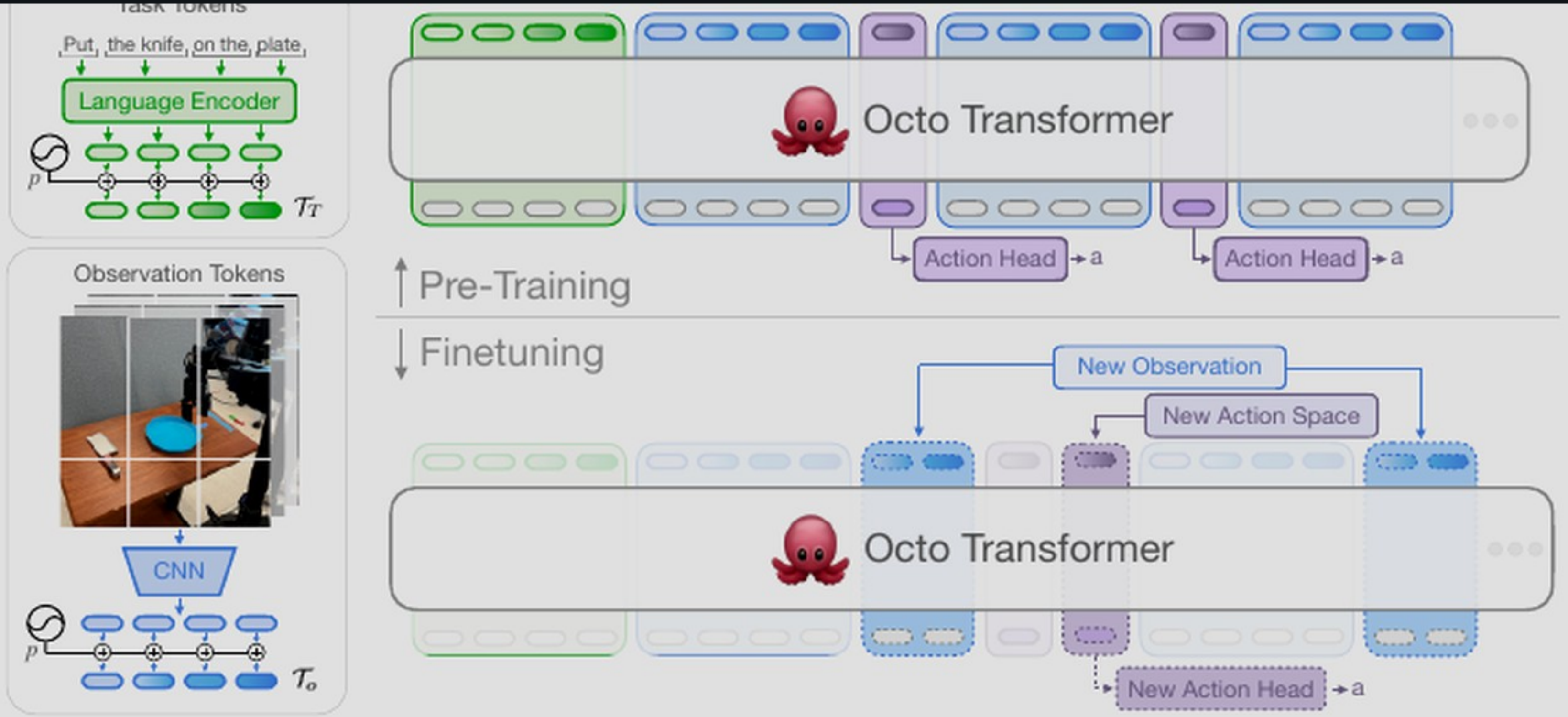


Fig. 2: **Model architecture. Left**: Octo tokenizes task descriptions (green) and input observations (blue) using a pretrained language model and a lightweight CNN, respectively. **Top**: The transformer backbone processes the sequence of task and observation tokens and produces readout tokens (purple) that get passed to output heads to produce actions. **Bottom**: The block-wise attention structure of the transformer allows us to add or remove inputs and outputs during finetuning: for example, we can add new observations (blue, dashed) or action spaces (purple, dashed) without modifying any pretrained parameters.

Octo's design is inspired by several recent advances in robot imitation learning and scalable transformer training, including the use of denoising diffusion objectives [34] for action decoding [17, 31, 85], the prediction of "action chunks", i.e., sequences of future actions [98, 17, 28], and model layouts and learning rate schedules inspired by the literature on scalable

we curate 800k for Octo training. We note that the RT-X model [67] used a more restricted subset of 350K episodes, so to the best of our knowledge, Octo is trained on the largest robotics manipulation demonstration dataset to date.

III. The Octo Model

12 actionable critiques · THE Reviewer talk figure

Say these aloud · ordered by how they change 2026 citation / BEHAVIOR planning

1 Absolute "first" TENTATIVE

Safer: first widely released open GRP + adapter FT on OXE-scale

2 RT-2-X parity task-limited

Mixed provenance — not "Octo ≈ 55B VLM" globally

3 Thin trials

~10 ZS / 20 FT — demand CIs / ≥3 seeds

4 Scratch not fully matched

Part of 52 pp gap may be objective/chunks

5 VC-1 = MSE MLP

Unfair vs multimodal diffusion — vision-features baseline only

6 Chunk T_p/T_a mushy

Main text "several"; only bimanual 64/12 — vs DP/π_0

7 Manual curation under-ablated

Which datasets drive 43→83%? Mix may dominate arch

8 Frozen T5 + lang poverty

Goals +25% = visual goals carry more bits

9 Wrist weakness (27%)

Undermines multi-cam marketing until coverage fixed

10 No mobile / nav

Cross-embodiment ≠ universal foundation model

11 93M vs 55B rhetoric

Easy-task parity ≠ equal representation power

12 Optimal demos only

No offline RL on suboptimal OXE

Reporting checklist · pair every headline %

Kanta: never let a bare percentage leave the room without its context

72%

Table I FT average

~100 demos · 50k steps · full update · ~5 h A5000 24 GB · 20 trials/domain

+29%

Zero-shot vs RT-1-X

Language · selected in-distribution WidowX/UR5/RT-1 tasks · ~10 trials/task

83%

Ablation aggregate

WidowX only · Octo-Small · 40 trials / 4 tasks (lang+goal) — not Base, not 9-robot

5%

Novel skill (Table VII)

Flip cup / insert block — cite whenever someone says “generalist zero-shot”

Also: 83/35/18 = diffusion / MSE / discrete · 27M/93M sizes · wrist 27% · lang 56%

Generalization honesty · Table VII WidowX

In-distribution	Novel objects	Novel environment	Novel skill
85%	**80%**	**40%**	**5%**
			flip / insert ANTI-HYPE

Cite 5% novel skill whenever someone says “generalist zero-shot”

Objects transfer; environments degrade; skills near floor — prevents foundation-model overclaim

Underspecified horizons · motion FLAG

Main text: “several” future actions · only concrete: bimanual pred 64 / exec 12

System	Chunk object	Generative	Ensemble?	Typical commit
ACT	k×14 joints	CVAE	Yes TE	ensembled @ 50 Hz
Diffusion Policy	Tp actions	DDPM/DDIM	warm / Ta	Ta≈8 @ ~10 Hz
Octo (this)	action chunk	DDPM readout	No (paper)	receding prefix
π_0 / $\pi_{0.5}$	H actions	Flow matching	No	open-loop K @ 20–50
BEHAVIOR’25	H≈30–32	FM + corr.	soft inpaint	~30 Hz receding

Empiricist: always log (H_pred, H_exec, Hz, diff_steps) · commit_seconds = H_exec/Hz · TE not dead globally

Citation guidance · how Kanta should talk about Octo

USE Octo as…

- ✓ Canonical open OXE GRP + finetune-first API
- ✓ Evidence diffusion chunks ≫ MSE/bins at multi-embod. scale
- ✓ Case study: mixture curation ≈ architecture
- ✓ Ancestor bridge into π_0 / OpenPI / BEHAVIOR stacks
- ✓ Operationalizable FT recipe (~100 demos / 24 GB)

AVOID using Octo as…

- ✗ Proof that open 93M ≈ closed 55B VLA generally
- ✗ Evidence that TE is obsolete in all regimes
- ✗ Fully solved "any robot, any sensor" FM
- ✗ Drop-in 2026 BEHAVIOR baseline
- ✗ Prefer $\pi_{0.5}$ / N1.7 lineage for production

VERDICT

Strong RSS 2024 open systems paper between ACT/DP and π_0/OpenVLA.
Praise openness + FT protocol; punish zero-shot overclaim and underspecified horizons.

Octo = infrastructure history + eval baseline · NOT 2026 winner weights · NOT BEHAVIOR drop-in scratch as well as state-of-the-art pretrained visual representations. Each domain uses ~ 100 target demonstrations and the same finetuning hyperparameters. In each domain, success rates are averaged over 20 trials. $*$: New observation input (force-torque proprioception). $\dagger$: New action space (joint position control).

placing, wiping a table with a cloth, and opening and closing drawers. For each robot, we selected two language tasks from the corresponding OXE dataset and performed 10 trials per task with varying initial conditions (details in Appendix F). The chosen tasks are "in-distribution" from the pre-training data, but the evaluation requires methods to generalize to new object positions, lighting conditions, backgrounds, and distractor objects. While all methods acted reasonably across tasks in the pretraining environments, we found that on average Octo had a 29% higher success rate than RT-1-X (35M parameters). For the WidowX and RT-1 Robot evaluations, we also compared to RT-2-X (55 billion parameters) [103] and found that Octo performed similarly.

Additionally, while RT-1-X and RT-2-X only support conditioning on language instructions, Octo also supports conditioning on goal images. We evaluated our model on the WidowX tasks using goal image conditioning and found that it achieved a 25% higher success rate than when evaluated with language conditioning. This is likely because goal images provide more information about how to achieve the task. In the BridgeV2 domain, we performed a fine-grained analysis of the zero-shot

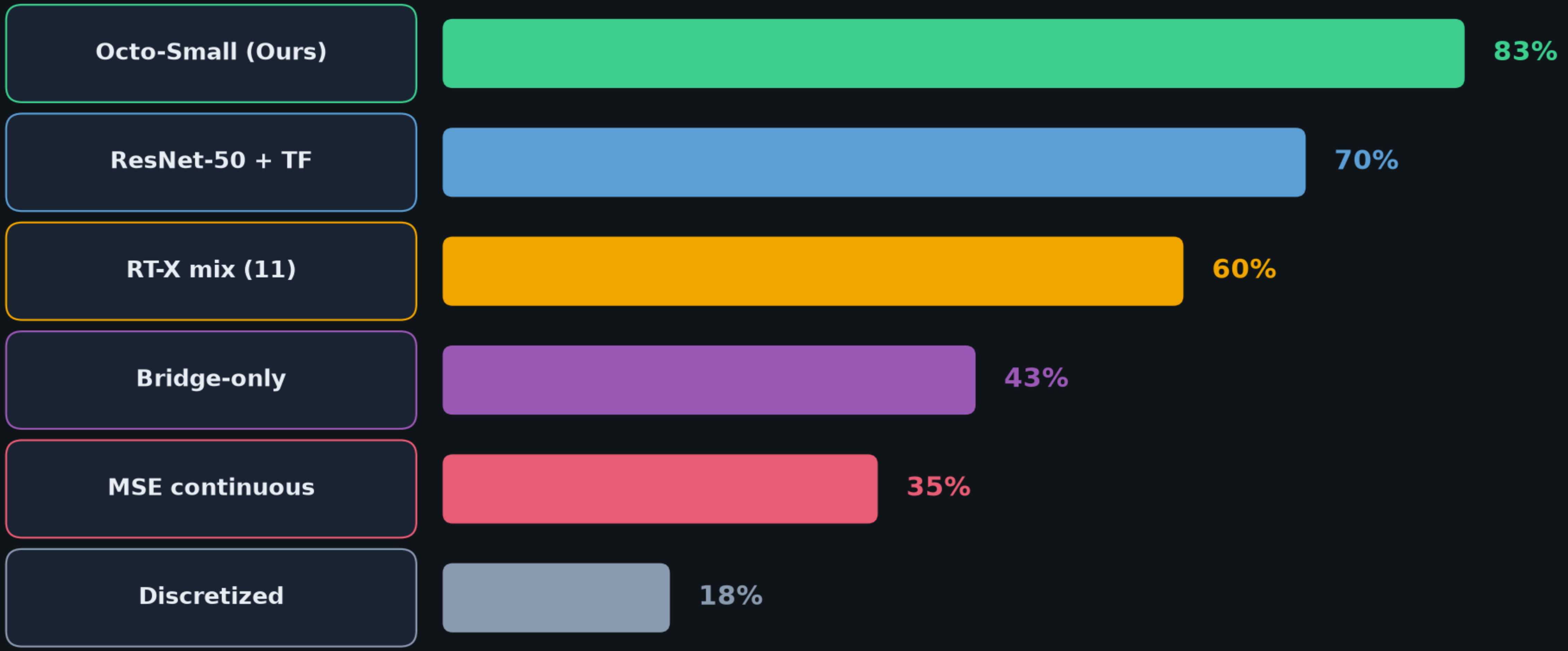
		Aggregate Performance
	Octo-Small (Ours)	**83%**
DATA	RT-X dataset mix [67]	60%
	Single robot dataset (Bridge Data)	43%
POLICY	Discretized Action Prediction [67]	18%
	Continuous Action Prediction (MSE)	35%
ARCH	Resnet-50 + Transformer[67]	70%

TABLE II: **Model Ablations.** We achieve best performance when using the ViT architecture, diffusion action head, and wide training data mixture. All evaluations are performed on the WidowX setup. Success rates are averaged over 40 trials across two language-conditioned tasks and two goal-conditioned tasks.

C. Design Decisions for Generalist Robot Policy Training

We have demonstrated the effectiveness of Octo as a zero-shot multi-robot controller and as an initialization for policy finetuning. We next analyze the effects of different design

Ablation board · Table II WidowX Octo-Small



Diffusion ≫ MSE/discrete · mix width 43→60→83 · ViT-first helps at pretrain scale

Strong RSS 2024 open systems paper between ACT/DP and π_0/OpenVLA

Praise openness + FT protocol
Punish zero-shot overclaim + underspecified horizons

Q&A · keep the spine clear

Octo: An Open-Source Generalist Robot Policy

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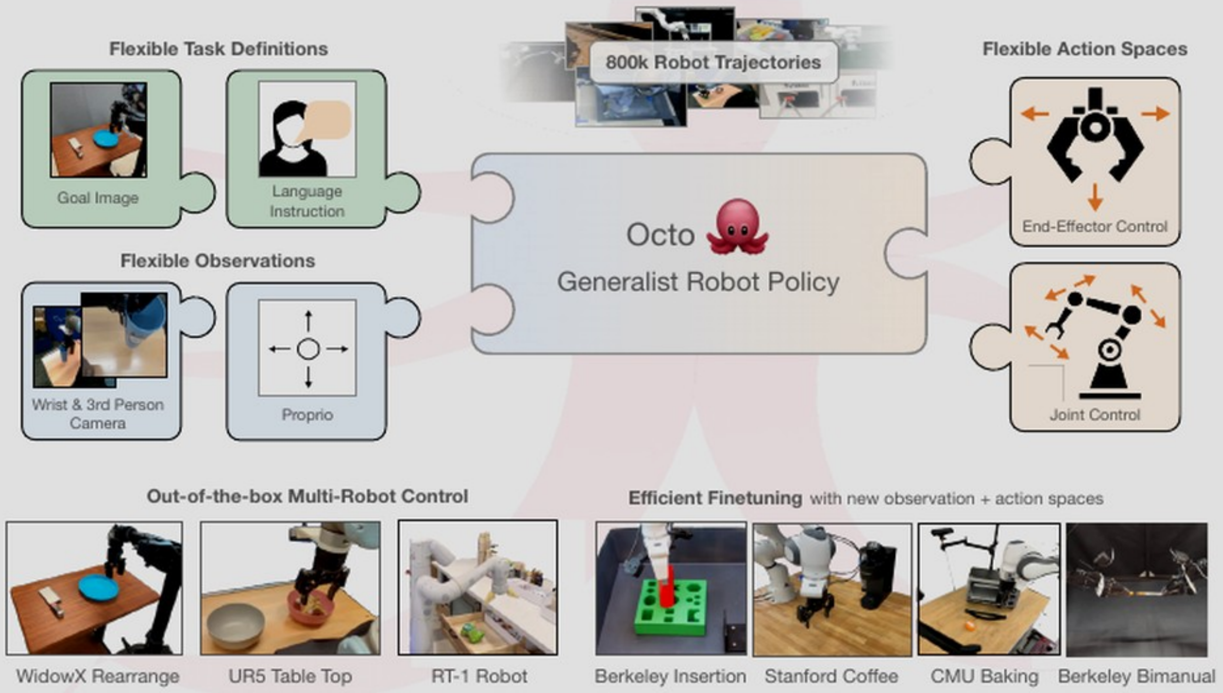

Fig. 1: We introduce Octo, an open-source, generalist policy for robotic manipulation. Octo is a transformer-based policy pretrained on 800k diverse robot episodes from the Open X-Embodiment dataset [67]. It supports flexible task and observation definitions and can be quickly finetuned to new observation and action spaces.

***Abstract*—Large policies pretrained on diverse robot datasets have the potential to transform robotic learning: instead of**

experiments across 9 robotic platforms, we demonstrate that Octo serves as a versatile policy initialization that can be effectively